

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems that can perform tasks that normally require human intelligence. These tasks include understanding language, recognizing images, making decisions, and solving problems.

Machine Learning (ML) is a part of AI in which computers learn patterns from data instead of being programmed with every rule. A machine learning model can be trained using examples and can then make predictions on new data.

Natural Language Processing (NLP) helps computers understand and process human language. NLP is used in chatbots, translation systems, sentiment analysis, and voice assistants.

Generative AI is a type of AI that can create new content such as text, images, code, or summaries. Large language models are commonly used for generating and understanding text.

AI applications are used in many areas, including education, healthcare, banking, transportation, customer service, and business analytics. The goal of these systems is to automate useful tasks and help people make better decisions.